

Relation Algebra

§10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$$\begin{array}{ll} \exists_R \square = R\square, & \exists_R = \exists_R \square \\ \mathbb{1} \sqsubseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ \exists_R = \mathbb{1} & \exists_R \text{ is } \mathbf{straight} \end{array}$$

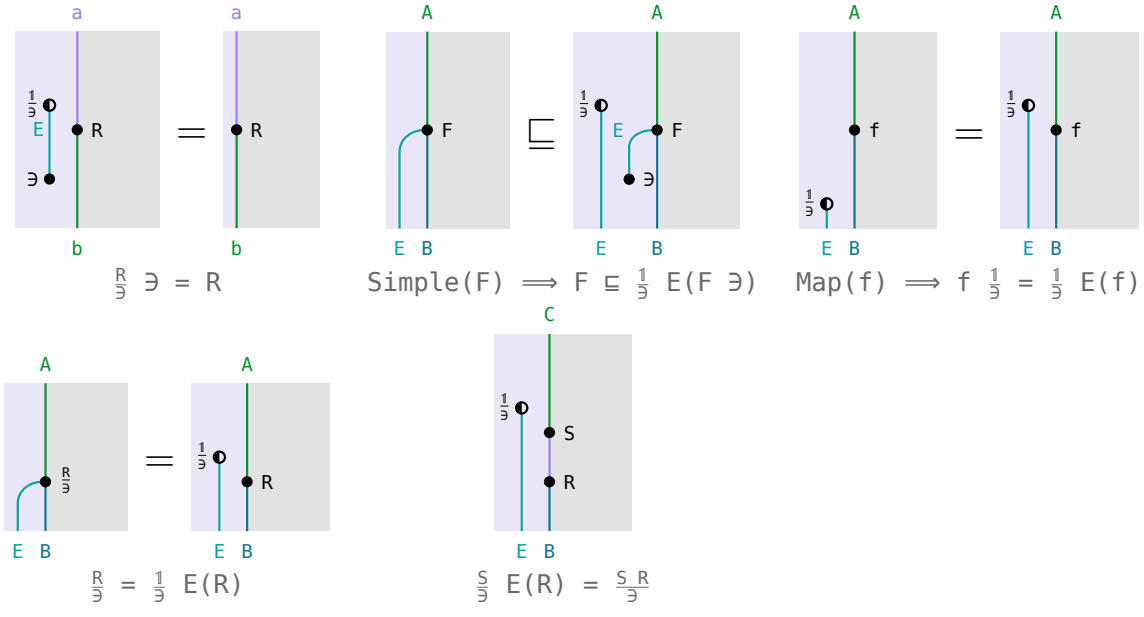
$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : PB \rightarrow B$, dropping the subscript.

$$\in \triangleq \exists^\circ : A \rightarrow PA$$

(10b)

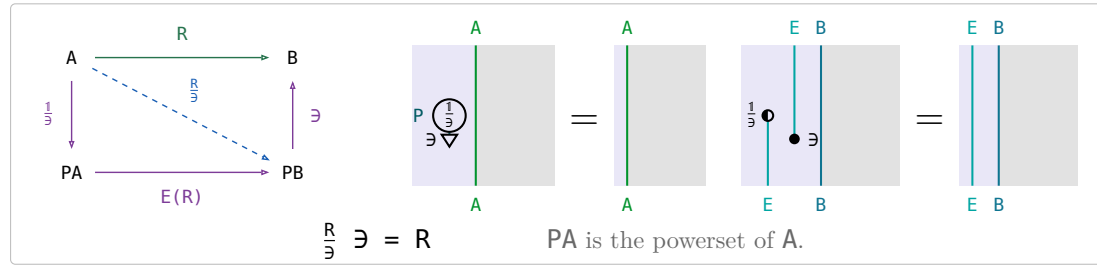
	law	the reading
1	$\exists_R \square = R\square, \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2	$\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3	$\frac{R}{\exists} : A \rightarrow PB$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4	$\text{Map}(\frac{R}{\exists})$	
5	$\frac{R}{\exists} \exists = R$	reading the chosen set back through $\exists$ returns the relation.
6	$\text{Simple}(F) \Rightarrow F \sqsubseteq \frac{F \exists}{\exists}$	A partial choice of sets is inside the total one.
7	the singleton map is monic: $\frac{1}{\exists} (\frac{1}{\exists})^\circ \sqsubseteq \mathbb{1}$ two points with the same one-person set are the same point.	The one-person set.
8	$\frac{\exists}{\exists} = \mathbb{1}$	Make the set of a set, then read it back one level down.
9	fusion: $\text{Map}(f) \Rightarrow \frac{f R}{\exists} = f \frac{R}{\exists}$	Naturality of the unit, $\text{Map}(f) \Rightarrow f \frac{1}{\exists} = \frac{1}{\exists} E(f)$. renaming a point and then taking its one-person set is taking the set and renaming inside it.
10	$\text{Map}(f) \wedge \mathbb{1}\square = \exists\square \Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}$	Rename first or take singletons first — the fusion row above at $R=1$.
11	$R\square = \exists\square \Rightarrow R /_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$	x and y match when R sends x and S sends y to the same set.
12	$E(R) \triangleq \frac{\exists R}{\exists}$	$E(R) : PA \rightarrow PB$, image of a set of A
13	$\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14	$\frac{S}{\exists} E(R) = \frac{S R}{\exists}$	absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{S R}{\exists}$, §Section 10.3
15	$E(R) = E(\frac{R}{\exists})$ union	the image of a set under R is the union of the R -sets of its points.
16	$\supseteq \triangleq \exists / \exists$	$xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$
17	$\subseteq = \supseteq^\circ$	$xs \subseteq ys \iff ys \supseteq xs$
18	$T^\circ \frac{T}{\exists} \sqsubseteq \in$	every T -value of a point is a member of its T -set: $\wedge(T)$ is a map and $\wedge(T)\exists = T$.

(10c)



§10.1. $\mathbf{i}\text{-}\mathbf{E}$ Power Allegory defined as adjunction

(10.1a)



$\mathbf{i}$ is the inclusion $\mathbf{Map}(\mathcal{A}) \rightarrow \mathcal{A}$, doing nothing, so $\mathbf{E}$ is both the functor and the monad $\mathbf{i}\mathbf{E}$.

§10.2. $\epsilon \setminus$ as a composite

$\epsilon \setminus Z = \subseteq (\frac{Z^\circ}{\exists})^\circ$ (10.2a)

$\mathbf{xS}$ is related to $\mathbf{c}$ by $\epsilon \setminus Z$ exactly when $\mathbf{xS} \subseteq \Lambda(Z^\circ)(\mathbf{c})$, the set of every $\mathbf{a}$ with $\mathbf{a} Z \mathbf{c}$

1	$\epsilon \setminus Z$	$Z = \epsilon \Lambda(Z^\circ)^\circ \text{ — } \frac{R}{\exists} \ni = R$
2	$= \epsilon \setminus (\epsilon (\frac{Z^\circ}{\exists})^\circ)$	$X \setminus (Y f^\circ) = (X \setminus Y) f^\circ, f \text{ a map}$
3	$= \subseteq (\frac{Z^\circ}{\exists})^\circ$	

$R \setminus S = \frac{R^\circ}{\exists} \subseteq (\frac{S^\circ}{\exists})^\circ$ (10.2b)

$\mathbf{b}$ is related to $\mathbf{c}$ by $R \setminus S$ exactly when $\Lambda(R^\circ)(\mathbf{b}) \subseteq \Lambda(S^\circ)(\mathbf{c})$, the R -preimage of $\mathbf{b}$ inside the S -preimage of $\mathbf{c}$

1	$R \setminus S$	$R = \epsilon \Lambda(R^\circ)^\circ \text{ — } \frac{R}{\exists} \ni = R$
2	$= (\epsilon (\frac{R^\circ}{\exists})^\circ) \setminus S$	$(XY) \setminus S = Y \setminus (X \setminus S)$
3	$= (\frac{R^\circ}{\exists})^\circ \setminus (\epsilon \setminus S)$	$f^\circ \setminus X = fX; \epsilon \setminus S = \subseteq \Lambda(S^\circ)^\circ \text{ — } \epsilon \setminus Z = \subseteq (\frac{Z^\circ}{\exists})^\circ$
4	$= \frac{R^\circ}{\exists} \subseteq (\frac{S^\circ}{\exists})^\circ$	

§10.3. $\mathcal{A} \cong \text{Kleisli}(E)$

(10.3a)

$$E(R) \triangleq \frac{\exists}{\exists} R, \quad \frac{1}{\exists} : A \rightarrow PA, \quad \text{union} = E(\exists) : PPA \rightarrow PA$$

$$f \diamond g \triangleq f \circ E(g) \circ \text{union}, \quad \text{for } f : A \rightarrow PB \text{ and } g : B \rightarrow PC$$

the monad is on $\text{Map}(\mathcal{A})$, not on the allegory: E is a relator on all relations, but $\frac{1}{\exists}$, union and $f \circ E(g) \circ \text{union}$ are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{aligned} \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} & \xrightarrow{\text{definition of } \diamond, \text{ union}=E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y)=E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\exists\right)} \\ & \xrightarrow{\frac{R}{\exists} \exists = R, \frac{R}{\exists} \exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{\frac{S}{\exists} E(R) = \frac{S}{\exists} R, \text{ absorption}} \boxed{\frac{SR}{\exists}} \end{aligned}$$

the first three steps only unfold $\diamond$ — E is a functor, so $E\left(\frac{R}{\exists}\right)\text{union}=E\left(\frac{R}{\exists}\exists\right)=E(R)$ and the union is gone.

Absorption, $\frac{S}{\exists} E(R) = \frac{S}{\exists} R$, is the whole law, and it is the functoriality of $\frac{\cdot}{\exists}$.

1 goes to $\frac{1}{\exists}$, the Kleisli identity, by definition — so $\frac{\cdot}{\exists}$ is an isomorphism of categories $\mathcal{A} \cong \text{Kleisli}(E)$, and §Section 10.1's $i \dashv E$ is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.